

Main Analysis: All Studies

STAT778 Project — gcKrig Robustness Simulation

Chengyan (Kurt) Ji

2026-05-04

Contents

1 Purpose	2
2 Setup	2
3 Study 1: K13A-GRID Reproduction	3
3.1 Load and tidy	3
3.2 Convergence	4
3.3 Runtime	4
3.4 MAE per cell	5
3.5 Bias summary	6
3.6 Numerical summary of MAE vs sample size	6
3.7 Average sample-size effect by true range	9
3.8 Headline figure: MAE vs sample size, faceted by range	9
4 Study 2: MISSPEC (Marginal Misspecification)	12
4.1 Load fit results	12
4.2 Build long-format table with truth attached	12
4.3 Convergence and runtime	13
4.4 Headline MAE table	14
4.5 GQT log-likelihood pathology under NegBin -> Poisson	15
4.6 Boxplot of estimates (the report-ready figure)	15
5 Study 3: MISSING (MCAR Robustness)	16
5.1 Load and tidy	16
5.2 MAE and bias by rate	16
5.3 Runtime by missingness rate	18
5.4 Figure: MAE vs missingness rate	18
6 Study 4: IRREGULAR (Spatial Layout)	19
6.1 Load and tidy	19
6.2 MAE and bias by layout	20
6.3 Figure: layouts side-by-side, then MAE bars	21
7 Study 5: ZIP (Zero-Inflation)	22
7.1 Load both well-specified and misspecified fits	23
7.2 Convergence	23
7.3 MAE: well-specified vs misspecified, side by side	24
7.4 Figure: MAE vs zero-inflation level	25
7.5 GQT instability at $p_0 = 0.4$	26

8	Headline Numbers for the Report	26
9	Files Produced	29
10	Session Info	29

1 Purpose

This document is the consolidated analysis layer for the project. It loads the fit results from all five completed studies and produces every table and figure cited in the written report:

- **K13A-GRID**: reproduction across the 18-cell parameter grid.
- **MISSPEC**: 2x2 factorial of marginal misspecification.
- **MISSING**: MCAR missingness at five rates.
- **IRREGULAR**: three spatial layouts (grid, random, clustered).
- **ZIP**: zero-inflation with correct vs misspecified fitting.

The per-study walkthrough Rmds (`k13a_grid_analysis.Rmd`, `misspec_analysis.Rmd`, etc.) remain the source of truth for detailed inspection and debugging. This file is the report-facing summary: every numerical claim and every figure in the report is generated here, so the report can be regenerated by re-knitting this single document.

2 Setup

```
suppressPackageStartupMessages({
  library(gcKrig)
})

source("code/R/logging.R")
source("code/R/io.R")
source("code/R/fit_core.R")

# Output directories for tables and figures
dir.create("visualization/report", recursive = TRUE, showWarnings = FALSE)
dir.create("tables/report", recursive = TRUE, showWarnings = FALSE)

# Generic tidier: one row per (scenario, replicate, method, parameter).
# Each study supplies its own extractor for scenario-level metadata
# (e.g., missing rate, layout type) via the `extra_cols` function.
tidy_fit <- function(f, extra_cols = function(cfg) data.frame()) {
  r <- f$results
  cfg <- f$scenario_config
  rows <- list()
  for (i in seq_len(nrow(r))) {
    if (!isTRUE(r$converged[i])) {
      rows[[length(rows) + 1]] <- cbind(
        scenario_id = f$scenario_id,
        extra_cols(cfg),
        replicate = r$replicate[i],
        method = r$method[i],
        converged = FALSE,
        runtime_sec = r$runtime_sec[i],
        parameter = NA_character_,
        estimate = NA_real_,

```

```

    stringsAsFactors = FALSE
  )
  next
}
est <- r$estimates[[i]]
if (is.null(est) || length(est) == 0) next
for (pname in names(est)) {
  rows[[length(rows) + 1]] <- cbind(
    scenario_id = f$scenario_id,
    extra_cols(cfg),
    replicate   = r$replicate[i],
    method      = r$method[i],
    converged    = TRUE,
    runtime_sec  = r$runtime_sec[i],
    parameter    = pname,
    estimate     = as.numeric(est[[pname]]),
    stringsAsFactors = FALSE
  )
}
}
do.call(rbind, rows)
}

# Round a numeric column to fixed digits, returning a character vector
# suitable for inclusion in printed tables. Avoids R's default
# scientific notation surprises in MAE/bias columns.
fmt <- function(x, digits = 3) formatC(x, format = "f", digits = digits)

# Save a data frame as a CSV in tables/report and also return it
# invisibly so chunk output remains usable.
save_table <- function(df, name) {
  path <- file.path("tables/report", paste0(name, ".csv"))
  write.csv(df, path, row.names = FALSE)
  invisible(df)
}

```

3 Study 1: K13A-GRID Reproduction

The 18-cell factorial: $n \in \{25, 49\}$, $\text{range} \in \{0.1, 0.3, 0.5\}$, $\lambda \in \{1, 5, 10\}$. 200 replications per cell, both GHK and GQT. This study anchors the report by establishing how the two methods behave under well-specified conditions.

3.1 Load and tidy

```

DATASET_K13A <- "K13A-GRID"
ids_k13a <- list_scenarios(DATASET_K13A, dir = "data/generated")
fits_k13a <- lapply(ids_k13a, function(id) load_fit_result(DATASET_K13A, id))
names(fits_k13a) <- ids_k13a

# Sanity check: 18 cells.
stopifnot(length(fits_k13a) == 18)

```

```

extra_k13a <- function(cfg) data.frame(
  n_total = cfg$layout$n_side^2,
  range = cfg$corr$range,
  lambda = cfg$marginal$lambda,
  stringsAsFactors = FALSE
)

long_k13a <- do.call(rbind, lapply(fits_k13a, tidy_fit, extra_cols = extra_k13a))

# Attach truth (Poisson intercept is on log scale).
long_k13a$truth <- NA_real_
is_int <- long_k13a$parameter %in% "Intercept"
is_rng <- long_k13a$parameter %in% "range"
long_k13a$truth[is_int] <- log(long_k13a$lambda[is_int])
long_k13a$truth[is_rng] <- long_k13a$range[is_rng]
long_k13a$abs_err <- abs(long_k13a$estimate - long_k13a$truth)
long_k13a$err <- long_k13a$estimate - long_k13a$truth

```

3.2 Convergence

```

conv_k13a <- aggregate(
  converged ~ method,
  data = unique(long_k13a[, c("scenario_id", "replicate", "method", "converged")]),
  FUN = mean
)
conv_k13a$conv_pct <- 100 * conv_k13a$converged
print(conv_k13a, row.names = FALSE)

## method converged conv_pct
## GHK 1.0000000 100.00000
## GQT 0.9994444 99.94444

save_table(conv_k13a, "k13a_convergence")

```

3.3 Runtime

```

rt_k13a <- unique(long_k13a[long_k13a$converged,
  c("scenario_id", "replicate", "method",
    "n_total", "runtime_sec")])
runtime_k13a <- aggregate(runtime_sec ~ n_total + method, data = rt_k13a, FUN = mean)
runtime_wide <- reshape(runtime_k13a, idvar = "n_total", timevar = "method",
  direction = "wide")
runtime_wide$ratio_GHK_to_GQT <- runtime_wide$runtime_sec.GHK /
  runtime_wide$runtime_sec.GQT
print(runtime_wide, row.names = FALSE, digits = 3)

## n_total runtime_sec.GHK runtime_sec.GQT ratio_GHK_to_GQT
## 25 0.270 0.0316 8.56
## 49 0.698 0.0846 8.25

save_table(runtime_wide, "k13a_runtime")

cat(sprintf(
  "\nOverall GHK-to-GQT speedup: %.2fx (n=25), %.2fx (n=49)\n",

```

```
runtime_wide$ratio_GHK_to_GQT[runtime_wide$n_total == 25],
runtime_wide$ratio_GHK_to_GQT[runtime_wide$n_total == 49]
))
```

```
##
```

```
## Overall GHK-to-GQT speedup: 8.56x (n=25), 8.25x (n=49)
```

3.4 MAE per cell

```
mae_k13a <- aggregate(
  abs_err ~ n_total + range + lambda + method + parameter,
  data = long_k13a[long_k13a$converged, ],
  FUN = mean
)
names(mae_k13a)[6] <- "MAE"

# Wide form for direct comparison.
mae_k13a_wide <- reshape(
  mae_k13a,
  idvar = c("n_total", "range", "lambda", "parameter"),
  timevar = "method", direction = "wide"
)
mae_k13a_wide$pct_diff <- 100 * (mae_k13a_wide$MAE.GQT - mae_k13a_wide$MAE.GHK) /
  mae_k13a_wide$MAE.GHK

# Sort and display
mae_k13a_wide <- mae_k13a_wide[order(mae_k13a_wide$parameter,
  mae_k13a_wide$n_total,
  mae_k13a_wide$range,
  mae_k13a_wide$lambda), ]
print(mae_k13a_wide, row.names = FALSE, digits = 3)
```

```
##  n_total range lambda parameter MAE.GHK MAE.GQT pct_diff
##      25   0.1      1 Intercept  0.1943  0.1896 -2.4000
##      25   0.1      5 Intercept  0.0860  0.0854 -0.6377
##      25   0.1     10 Intercept  0.0582  0.0579 -0.4262
##      25   0.3      1 Intercept  0.3207  0.2889 -9.9204
##      25   0.3      5 Intercept  0.1631  0.1645  0.8285
##      25   0.3     10 Intercept  0.1132  0.1133  0.0980
##      25   0.5      1 Intercept  0.5196  0.3458 -33.4443
##      25   0.5      5 Intercept  0.2039  0.2053  0.6961
##      25   0.5     10 Intercept  0.1345  0.1337 -0.5955
##      49   0.1      1 Intercept  0.1732  0.1671 -3.5609
##      49   0.1      5 Intercept  0.0724  0.0730  0.7992
##      49   0.1     10 Intercept  0.0573  0.0574  0.0673
##      49   0.3      1 Intercept  0.3103  0.2693 -13.2085
##      49   0.3      5 Intercept  0.1537  0.1550  0.8197
##      49   0.3     10 Intercept  0.1016  0.1027  1.1078
##      49   0.5      1 Intercept  0.4129  0.3550 -14.0165
##      49   0.5      5 Intercept  0.2034  0.2006 -1.3644
##      49   0.5     10 Intercept  0.1396  0.1399  0.2101
##      25   0.1      1      range  0.0972  0.1085 11.6411
##      25   0.1      5      range  0.1010  0.1000 -0.9990
##      25   0.1     10      range  0.0951  0.0948 -0.3834
```

```
##      25  0.3    1    range 0.1632 0.1596 -2.2192
##      25  0.3    5    range 0.1192 0.1155 -3.1119
##      25  0.3   10    range 0.1314 0.1299 -1.1529
##      25  0.5    1    range 0.2207 0.2077 -5.9024
##      25  0.5    5    range 0.1581 0.1535 -2.9177
##      25  0.5   10    range 0.1581 0.1565 -0.9587
##      49  0.1    1    range 0.0792 0.0639 -19.3297
##      49  0.1    5    range 0.0775 0.0832  7.3712
##      49  0.1   10    range 0.0773 0.0786  1.6657
##      49  0.3    1    range 0.0997 0.0970 -2.7270
##      49  0.3    5    range 0.0699 0.0669 -4.2944
##      49  0.3   10    range 0.0605 0.0599 -1.1113
##      49  0.5    1    range 0.1555 0.1412 -9.2175
##      49  0.5    5    range 0.1064 0.1019 -4.1968
##      49  0.5   10    range 0.1092 0.1075 -1.4984
```

```
save_table(mae_k13a_wide, "k13a_mae_wide")
```

```
cat(sprintf("\nMax absolute %% MAE difference between methods: %.1f%%\n",
           max(abs(mae_k13a_wide$pct_diff), na.rm = TRUE)))
```

```
##
## Max absolute % MAE difference between methods: 33.4%
```

3.5 Bias summary

Bias (mean signed error) per cell. Useful for confirming that the estimators are approximately unbiased and that any MAE growth with range or low-lambda is variance-driven rather than bias-driven.

```
bias_k13a <- aggregate(
  err ~ n_total + range + lambda + method + parameter,
  data = long_k13a[long_k13a$converged, ],
  FUN = mean
)
names(bias_k13a)[6] <- "bias"

# Compact summary: mean and max |bias| per (method, parameter).
bias_compact <- aggregate(
  bias ~ method + parameter, data = bias_k13a,
  FUN = function(x) c(mean = mean(x), max_abs = max(abs(x)))
)
bias_compact <- do.call(data.frame, bias_compact)
print(bias_compact, row.names = FALSE, digits = 3)

## method parameter bias.mean bias.max_abs
## GHK Intercept -0.04616      0.2213
## GQT Intercept -0.01824      0.0599
## GHK range     -0.01910      0.0495
## GQT range      0.00813      0.0854

save_table(bias_k13a, "k13a_bias")
```

3.6 Numerical summary of MAE vs sample size

```

# this table is the numerical version of the MAE-vs-n plots.
# each row fixes the parameter, true range, lambda, and method,
# then compares MAE at n = 25 and n = 49.

mae_vs_n <- mae_k13a[, c("parameter", "range", "lambda", "method",
                        "n_total", "MAE")]

mae_vs_n_wide <- reshape(
  mae_vs_n,
  idvar = c("parameter", "range", "lambda", "method"),
  timevar = "n_total",
  direction = "wide"
)

# make column names clearer for reporting
names(mae_vs_n_wide)[names(mae_vs_n_wide) == "range"] <- "true_range"
names(mae_vs_n_wide)[names(mae_vs_n_wide) == "MAE.25"] <- "MAE_n25"
names(mae_vs_n_wide)[names(mae_vs_n_wide) == "MAE.49"] <- "MAE_n49"

# change from n = 25 to n = 49
mae_vs_n_wide$change_49_minus_25 <- mae_vs_n_wide$MAE_n49 -
  mae_vs_n_wide$MAE_n25

# positive values mean MAE decreased when n increased
mae_vs_n_wide$pct_reduction <- 100 *
  (mae_vs_n_wide$MAE_n25 - mae_vs_n_wide$MAE_n49) /
  mae_vs_n_wide$MAE_n25

mae_vs_n_wide <- mae_vs_n_wide[
  order(mae_vs_n_wide$parameter,
        mae_vs_n_wide$true_range,
        mae_vs_n_wide$lambda,
        mae_vs_n_wide$method),
]

print(mae_vs_n_wide, row.names = FALSE, digits = 3)

```

##	parameter	true_range	lambda	method	MAE_n25	MAE_n49	change_49_minus_25
##	Intercept	0.1	1	GHK	0.1943	0.1732	-0.021045
##	Intercept	0.1	1	GQT	0.1896	0.1671	-0.022551
##	Intercept	0.1	5	GHK	0.0860	0.0724	-0.013557
##	Intercept	0.1	5	GQT	0.0854	0.0730	-0.012430
##	Intercept	0.1	10	GHK	0.0582	0.0573	-0.000861
##	Intercept	0.1	10	GQT	0.0579	0.0574	-0.000575
##	Intercept	0.3	1	GHK	0.3207	0.3103	-0.010386
##	Intercept	0.3	1	GQT	0.2889	0.2693	-0.019560
##	Intercept	0.3	5	GHK	0.1631	0.1537	-0.009391
##	Intercept	0.3	5	GQT	0.1645	0.1550	-0.009482
##	Intercept	0.3	10	GHK	0.1132	0.1016	-0.011580
##	Intercept	0.3	10	GQT	0.1133	0.1027	-0.010565
##	Intercept	0.5	1	GHK	0.5196	0.4129	-0.106746
##	Intercept	0.5	1	GQT	0.3458	0.3550	0.009164
##	Intercept	0.5	5	GHK	0.2039	0.2034	-0.000546
##	Intercept	0.5	5	GQT	0.2053	0.2006	-0.004740

##	Intercept	0.5	10	GHK	0.1345	0.1396	0.005162
##	Intercept	0.5	10	GQT	0.1337	0.1399	0.006256
##	range	0.1	1	GHK	0.0972	0.0792	-0.017927
##	range	0.1	1	GQT	0.1085	0.0639	-0.044556
##	range	0.1	5	GHK	0.1010	0.0775	-0.023511
##	range	0.1	5	GQT	0.1000	0.0832	-0.016789
##	range	0.1	10	GHK	0.0951	0.0773	-0.017783
##	range	0.1	10	GQT	0.0948	0.0786	-0.016130
##	range	0.3	1	GHK	0.1632	0.0997	-0.063536
##	range	0.3	1	GQT	0.1596	0.0970	-0.062633
##	range	0.3	5	GHK	0.1192	0.0699	-0.049337
##	range	0.3	5	GQT	0.1155	0.0669	-0.048628
##	range	0.3	10	GHK	0.1314	0.0605	-0.070870
##	range	0.3	10	GQT	0.1299	0.0599	-0.070027
##	range	0.5	1	GHK	0.2207	0.1555	-0.065188
##	range	0.5	1	GQT	0.2077	0.1412	-0.066496
##	range	0.5	5	GHK	0.1581	0.1064	-0.051669
##	range	0.5	5	GQT	0.1535	0.1019	-0.051522
##	range	0.5	10	GHK	0.1581	0.1092	-0.048887
##	range	0.5	10	GQT	0.1565	0.1075	-0.049008
##	pct_reduction						
##	10.832						
##	11.893						
##	15.765						
##	14.547						
##	1.480						
##	0.992						
##	3.239						
##	6.771						
##	5.757						
##	5.765						
##	10.231						
##	9.326						
##	20.544						
##	-2.650						
##	0.268						
##	2.308						
##	-3.838						
##	-4.680						
##	18.449						
##	41.073						
##	23.276						
##	16.789						
##	18.697						
##	17.024						
##	38.921						
##	39.238						
##	41.389						
##	42.104						
##	53.932						
##	53.912						
##	29.535						
##	32.017						
##	32.686						

```
##      33.573
##      30.930
##      31.307
```

```
save_table(mae_vs_n_wide, "k13a_mae_vs_n_wide")
```

This gives the exact numerical representation of each line in the plot below.

A compact summary averaging over λ :

3.7 Average sample-size effect by true range

```
# average the n = 25 to n = 49 improvement over lambda values.
# this summarizes the sample-size effect within each true range panel.
```

```
mae_n_effect_by_range <- aggregate(
  cbind(MAE_n25, MAE_n49, change_49_minus_25, pct_reduction) ~
    parameter + true_range + method,
  data = mae_vs_n_wide,
  FUN = mean
)
```

```
mae_n_effect_by_range <- mae_n_effect_by_range[
  order(mae_n_effect_by_range$parameter,
    mae_n_effect_by_range$true_range,
    mae_n_effect_by_range$method),
]
```

```
print(mae_n_effect_by_range, row.names = FALSE, digits = 3)
```

```
## parameter true_range method MAE_n25 MAE_n49 change_49_minus_25 pct_reduction
## Intercept      0.1   GHK  0.1128  0.1010          -0.01182          9.36
## Intercept      0.1   GQT  0.1110  0.0992          -0.01185          9.14
## Intercept      0.3   GHK  0.1990  0.1886          -0.01045          6.41
## Intercept      0.3   GQT  0.1889  0.1757          -0.01320          7.29
## Intercept      0.5   GHK  0.2860  0.2520          -0.03404          5.66
## Intercept      0.5   GQT  0.2283  0.2318           0.00356         -1.67
##      range      0.1   GHK  0.0978  0.0780          -0.01974         20.14
##      range      0.1   GQT  0.1011  0.0753          -0.02583         24.96
##      range      0.3   GHK  0.1380  0.0767          -0.06125         44.75
##      range      0.3   GQT  0.1350  0.0746          -0.06043         45.08
##      range      0.5   GHK  0.1789  0.1237          -0.05525         31.05
##      range      0.5   GQT  0.1726  0.1169          -0.05568         32.30
```

```
save_table(mae_n_effect_by_range, "k13a_mae_n_effect_by_range")
```

3.8 Headline figure: MAE vs sample size, faceted by range

```
plot_mae_panel <- function(mae_tab, param_name) {
  d <- mae_tab[mae_tab$parameter == param_name, ]
  ranges <- sort(unique(d$range))
  par(mfrow = c(1, length(ranges)),
      oma = c(0, 0, 3, 0), mar = c(4, 4, 3, 1))
  for (r in ranges) {
    dr <- d[d$range == r, ]
```

```

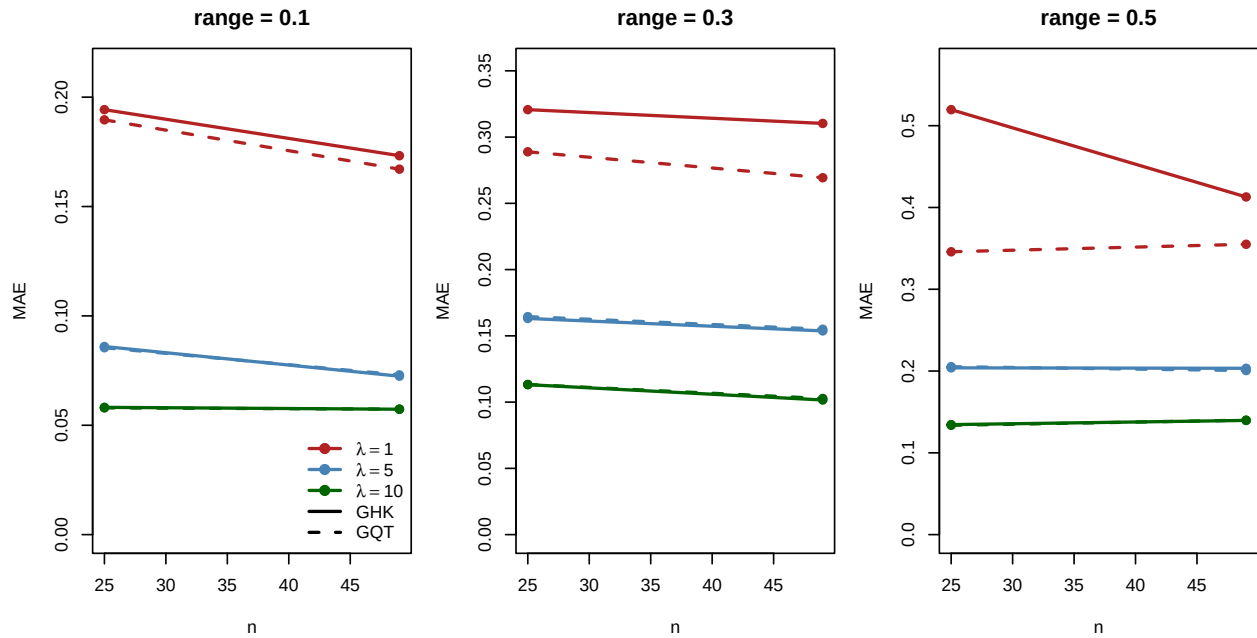
y_max <- max(dr$MAE, na.rm = TRUE) * 1.1
plot(NA, xlim = range(dr$n_total), ylim = c(0, y_max),
     xlab = "n", ylab = "MAE",
     main = sprintf("range = %g", r))
lams <- sort(unique(dr$lambda))
cols <- c("firebrick", "steelblue", "darkgreen")
ltys <- c(1, 2)
for (li in seq_along(lams)) {
  for (mi in seq_along(c("GHK", "GQT"))) {
    m <- c("GHK", "GQT")[mi]
    sub <- dr[dr$lambda == lams[li] & dr$method == m, ]
    sub <- sub[order(sub$n_total), ]
    lines(sub$n_total, sub$MAE,
          col = cols[li], lty = ltys[mi], lwd = 2)
    points(sub$n_total, sub$MAE, col = cols[li], pch = 19)
  }
}
if (r == ranges[1]) {
  legend("bottomright",
        legend = c(expression(lambda == 1),
                    expression(lambda == 5),
                    expression(lambda == 10),
                    "GHK", "GQT"),
        col = c("firebrick", "steelblue", "darkgreen", "black", "black"),
        lty = c(1, 1, 1, 1, 2),
        lwd = 2,
        pch = c(19, 19, 19, NA, NA),
        bty = "n",
        cex = 1)
}
}

mtext(sprintf("%s: MAE vs n (solid=GHK, dashed=GQT, colors=lambda)",
              param_name),
       outer = TRUE, cex = 1.1)
par(mfrow = c(1, 1))
}

plot_mae_panel(mae_k13a, "Intercept")

```

Intercept: MAE vs n (solid=GHK, dashed=GQT, colors=lambda)

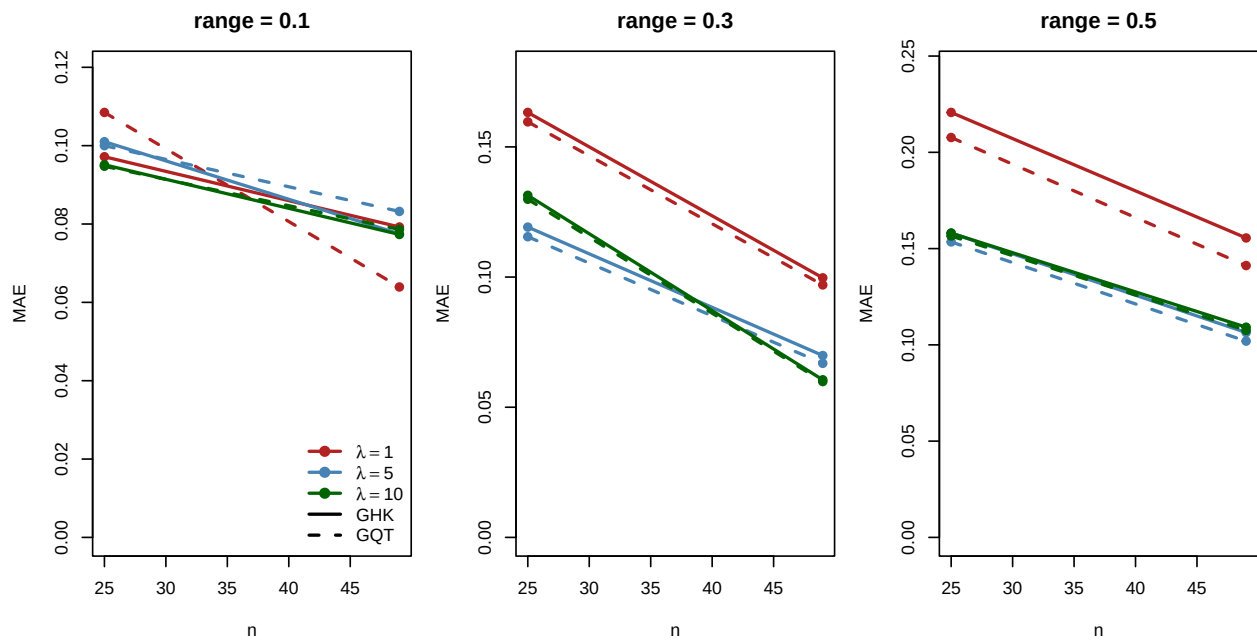


```
pdf("visualization/report/k13a_mae_intercept.pdf", width = 6, height = 5)
plot_mae_panel(mae_k13a, "Intercept")
dev.off()
```

```
## pdf
## 2
```

```
plot_mae_panel(mae_k13a, "range")
```

range: MAE vs n (solid=GHK, dashed=GQT, colors=lambda)



```
pdf("visualization/report/k13a_mae_range.pdf", width = 6, height = 5)
plot_mae_panel(mae_k13a, "range")
```

```
dev.off()
```

```
## pdf  
## 2
```

4 Study 2: MISSPEC (Marginal Misspecification)

A 2x2 factorial: generating family (Poisson or NegBin) crossed with fitting family (Poisson or NegBin), $n = 49$, $r = 0.3$, 30 replications per cell. Tests whether ignoring overdispersion or imposing it unnecessarily distorts estimation.

4.1 Load fit results

```
fit_dir_misspec <- file.path("data", "fits", "MISSPEC")  
fit_files_misspec <- list.files(fit_dir_misspec, pattern = "\\..rds$",  
                                full.names = TRUE)  
stopifnot(length(fit_files_misspec) == 4)  
fits_misspec <- lapply(fit_files_misspec, readRDS)  
names(fits_misspec) <- basename(fit_files_misspec)
```

4.2 Build long-format table with truth attached

```
truth_for_misspec <- function(pname, gen_cfg, fit_spec) {  
  p <- tolower(gsub("[[:space:]]+", "", pname))  
  gen_marg <- gen_cfg$marginal  
  if (grepl("range", p)) return(gen_cfg$corr$range)  
  if (grepl("kappa", p)) return(gen_cfg$corr$kappa)  
  if (grepl("nugget", p)) return(gen_cfg$corr$nugget)  
  if (grepl("^od$|overdisp|dispersion", p)) {  
    if (gen_marg$family == "negbin" && fit_spec$family == "negbin") {  
      return(gen_marg$od)  
    }  
    return(NA_real_)  
  }  
  # Intercept-scale mean  
  if (gen_marg$family == "poisson") return(log(gen_marg$lambda))  
  if (gen_marg$family == "negbin") return(gen_marg$beta[1])  
  NA_real_  
}  
  
flatten_misspec <- function(fr) {  
  r <- fr$results  
  rows <- list()  
  for (i in seq_len(nrow(r))) {  
    if (!isTRUE(r$converged[i])) next  
    est <- r$estimates[[i]]  
    if (is.null(est) || length(est) == 0) next  
    for (pname in names(est)) {  
      truth_val <- truth_for_misspec(pname, fr$scenario_config,  
                                     fr$fit_marginal_spec)  
      rows[[length(rows) + 1]] <- data.frame(  
        gen_family = fr$scenario_config$marginal$family,
```

```

    fit_family = fr$fit_marginal_spec$family,
    misspecified = fr$misspecified,
    cell_label = paste(fr$scenario_config$marginal$family, ">",
                       fr$fit_marginal_spec$family),
    replicate = r$replicate[i],
    method = r$method[i],
    runtime_sec = r$runtime_sec[i],
    loglik = r$loglik[i],
    parameter = pname,
    estimate = as.numeric(est[[pname]]),
    truth_val = truth_val,
    abs_err = if (is.na(truth_val)) NA_real_
               else abs(as.numeric(est[[pname]]) - truth_val),
    stringsAsFactors = FALSE
  )
}
}
do.call(rbind, rows)
}

long_misspec <- do.call(rbind, lapply(fits_misspec, flatten_misspec))

```

4.3 Convergence and runtime

```

# Reconstruct one-row-per-fit for convergence purposes.
fit_rows_misspec <- do.call(rbind, lapply(fits_misspec, function(fr) {
  r <- fr$results
  data.frame(
    cell_label = paste(fr$scenario_config$marginal$family, ">",
                      fr$fit_marginal_spec$family),
    method = r$method,
    converged = r$converged,
    runtime_sec = r$runtime_sec
  )
}))

conv_misspec <- aggregate(converged ~ cell_label + method,
                          data = fit_rows_misspec, FUN = mean)
print(conv_misspec, row.names = FALSE)

##           cell_label method converged
## negbin -> negbin      GHK          1
## negbin -> poisson     GHK          1
## poisson -> negbin     GHK          1
## poisson -> poisson    GHK          1
## negbin -> negbin      GQT          1
## negbin -> poisson     GQT          1
## poisson -> negbin     GQT          1
## poisson -> poisson    GQT          1

rt_misspec <- aggregate(runtime_sec ~ cell_label + method,
                        data = fit_rows_misspec[fit_rows_misspec$converged, ],
                        FUN = mean)
print(rt_misspec, row.names = FALSE, digits = 3)

```

```
##      cell_label method runtime_sec
##      negbin -> negbin      GHK      1.3448
##      negbin -> poisson     GHK      0.9330
##      poisson -> negbin     GHK      1.3319
##      poisson -> poisson    GHK      0.6581
##      negbin -> negbin      GQT      0.2015
##      negbin -> poisson     GQT      0.0925
##      poisson -> negbin     GQT      0.1782
##      poisson -> poisson    GQT      0.0879
```

4.4 Headline MAE table

```
mae_misspec <- aggregate(
  abs_err ~ cell_label + method + parameter,
  data = long_misspec[!is.na(long_misspec$abs_err), ],
  FUN = mean
)
names(mae_misspec)[4] <- "MAE"
mae_misspec <- mae_misspec[order(mae_misspec$parameter,
                                mae_misspec$cell_label,
                                mae_misspec$method), ]
print(mae_misspec, row.names = FALSE, digits = 3)
```

```
##      cell_label method      parameter      MAE
##      negbin -> negbin      GHK      Intercept 0.4047
##      negbin -> negbin      GQT      Intercept 0.3963
##      negbin -> poisson     GHK      Intercept 0.4447
##      negbin -> poisson     GQT      Intercept 0.4335
##      poisson -> negbin     GHK      Intercept 0.1611
##      poisson -> negbin     GQT      Intercept 0.1613
##      poisson -> poisson    GHK      Intercept 0.1575
##      poisson -> poisson    GQT      Intercept 0.1599
##      negbin -> negbin      GHK      overdispersion 0.3228
##      negbin -> negbin      GQT      overdispersion 0.4551
##      negbin -> negbin      GHK      range 0.1052
##      negbin -> negbin      GQT      range 0.1506
##      negbin -> poisson     GHK      range 0.2099
##      negbin -> poisson     GQT      range 0.2075
##      poisson -> negbin     GHK      range 0.1053
##      poisson -> negbin     GQT      range 0.1084
##      poisson -> poisson    GHK      range 0.0827
##      poisson -> poisson    GQT      range 0.0794
```

```
save_table(mae_misspec, "misspec_mae")
```

```
# Estimate summary so we can report range bias under NegBin -> Poisson.
```

```
est_summary_misspec <- aggregate(
  estimate ~ cell_label + method + parameter,
  data = long_misspec, FUN = mean
)
names(est_summary_misspec)[4] <- "mean_est"
print(est_summary_misspec[est_summary_misspec$cell_label == "negbin -> poisson", ],
      row.names = FALSE, digits = 3)
```

```
##           cell_label method parameter mean_est
## negbin -> poisson   GHK Intercept   1.6031
## negbin -> poisson   GQT Intercept   1.6270
## negbin -> poisson   GHK      range   0.0901
## negbin -> poisson   GQT      range   0.0925
```

4.5 GQT log-likelihood pathology under NegBin -> Poisson

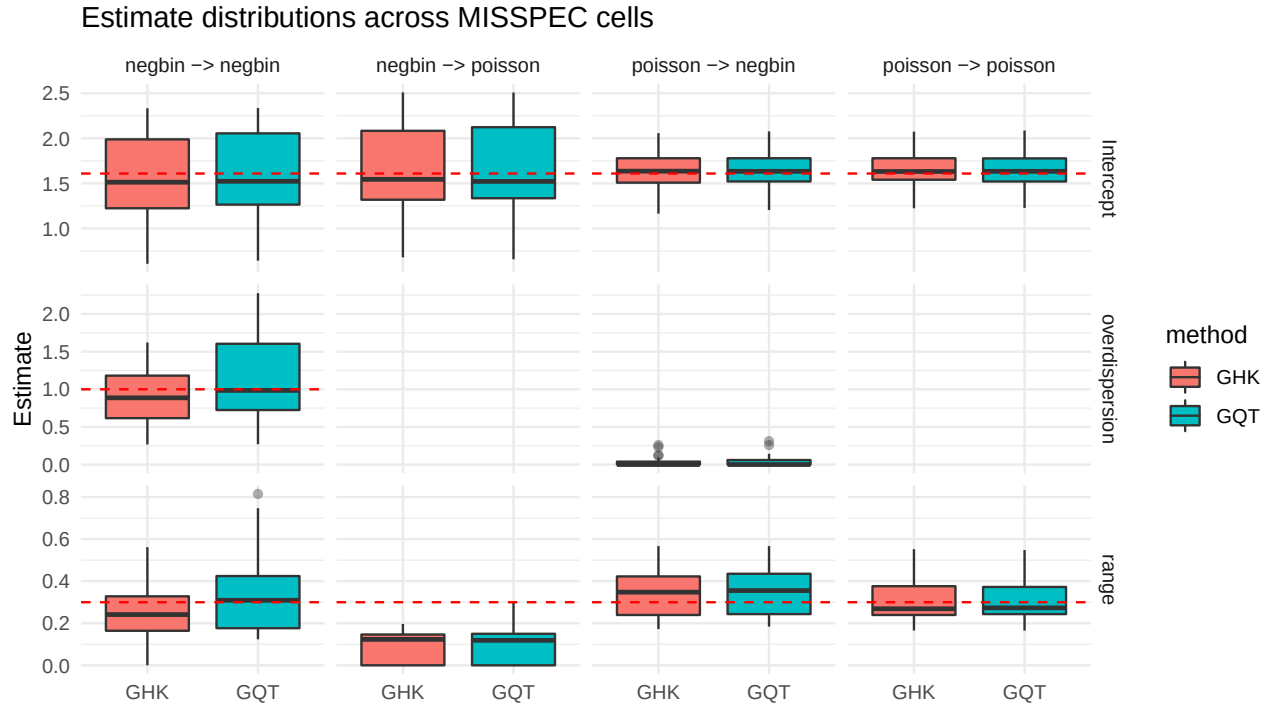
The GQT surrogate diverges numerically in this misspecification cell. We document it explicitly because AIC/BIC computed downstream from GQT cannot be trusted here even when the point estimates look fine.

```
ll_summary <- aggregate(
  loglik ~ cell_label + method,
  data = long_misspec[long_misspec$parameter == "Intercept", ],
  FUN = mean
)
print(ll_summary, row.names = FALSE)
```

```
##           cell_label method      loglik
## negbin -> negbin      GHK -1.206089e+02
## negbin -> poisson    GHK -1.558833e+02
## poisson -> negbin    GHK -9.499836e+01
## poisson -> poisson   GHK -9.523426e+01
## negbin -> negbin      GQT -1.200212e+02
## negbin -> poisson    GQT -3.177498e+13
## poisson -> negbin    GQT -9.484918e+01
## poisson -> poisson   GQT -9.515636e+01
```

4.6 Boxplot of estimates (the report-ready figure)

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  library(ggplot2)
  p_box <- ggplot(long_misspec,
    aes(x = method, y = estimate, fill = method)) +
    geom_boxplot(outlier.alpha = 0.4) +
    facet_grid(parameter ~ cell_label, scales = "free_y") +
    geom_hline(aes(yintercept = truth), color = "red",
      linetype = 2, na.rm = TRUE) +
    labs(x = NULL, y = "Estimate",
      title = "Estimate distributions across MISSPEC cells") +
    theme_minimal()
  print(p_box)
  ggsave("visualization/report/misspec_estimates.pdf",
    p_box, width = 10, height = 6)
}
```



5 Study 3: MISSING (MCAR Robustness)

Five MCAR rates $\{0, 0.1, 0.2, 0.3, 0.5\}$ on the Poisson(5) baseline, $n = 49$, $r = 0.3$, 200 replications per rate.

5.1 Load and tidy

```

DATASET_MISS <- "MISSING"
ids_miss <- list_scenarios(DATASET_MISS, dir = "data/generated")
fits_miss <- lapply(ids_miss, function(id) load_fit_result(DATASET_MISS, id))
names(fits_miss) <- ids_miss

extra_miss <- function(cfg) data.frame(
  missing_rate = cfg$missing_rate,
  stringsAsFactors = FALSE
)

long_miss <- do.call(rbind, lapply(fits_miss, tidy_fit, extra_cols = extra_miss))
long_miss$truth <- NA_real_
long_miss$truth[long_miss$parameter %in% "Intercept"] <- log(5)
long_miss$truth[long_miss$parameter %in% "range"] <- 0.3
long_miss$abs_err <- abs(long_miss$estimate - long_miss$truth)
long_miss$err <- long_miss$estimate - long_miss$truth

```

5.2 MAE and bias by rate

```

mae_miss <- aggregate(
  abs_err ~ missing_rate + method + parameter,
  data = long_miss[long_miss$converged, ],
  FUN = mean
)

```

```

)
names(mae_miss)[4] <- "MAE"

bias_miss <- aggregate(
  err ~ missing_rate + method + parameter,
  data = long_miss[long_miss$converged, ],
  FUN = mean
)
names(bias_miss)[4] <- "bias"

mae_miss_wide <- reshape(
  mae_miss, idvar = c("missing_rate", "parameter"),
  timevar = "method", direction = "wide"
)
print(mae_miss_wide, row.names = FALSE, digits = 3)

## missing_rate parameter MAE.GHK MAE.GQT
##      0.0 Intercept  0.1564  0.1585
##      0.1 Intercept  0.1608  0.1614
##      0.2 Intercept  0.1605  0.1619
##      0.3 Intercept  0.1593  0.1601
##      0.5 Intercept  0.1709  0.1689
##      0.0      range  0.0678  0.0660
##      0.1      range  0.0752  0.0729
##      0.2      range  0.0766  0.0761
##      0.3      range  0.0892  0.0867
##      0.5      range  0.1083  0.1066

save_table(mae_miss_wide, "missing_mae")

# Display bias (claim: stays near zero across all rates under MCAR).
bias_miss_wide <- reshape(
  bias_miss, idvar = c("missing_rate", "parameter"),
  timevar = "method", direction = "wide"
)
cat("\nBias by missingness rate:\n")

##
## Bias by missingness rate:

print(bias_miss_wide, row.names = FALSE, digits = 3)

## missing_rate parameter bias.GHK  bias.GQT
##      0.0 Intercept -0.01834 -1.84e-02
##      0.1 Intercept -0.02347 -2.25e-02
##      0.2 Intercept -0.02002 -2.02e-02
##      0.3 Intercept -0.01335 -1.40e-02
##      0.5 Intercept -0.02310 -2.09e-02
##      0.0      range -0.00232  1.92e-03
##      0.1      range -0.00572 -1.39e-04
##      0.2      range -0.00185  1.90e-03
##      0.3      range -0.00545 -1.84e-05
##      0.5      range -0.00525 -4.35e-04

```

```

save_table(bias_miss_wide, "missing_bias")

# Headline: MAE inflation from 0% to 50%
for (param in c("Intercept", "range")) {
  for (m in c("GHK", "GQT")) {
    base <- mae_miss$MAE[mae_miss$missing_rate == 0 &
                        mae_miss$parameter == param &
                        mae_miss$method == m]
    high <- mae_miss$MAE[mae_miss$missing_rate == 0.5 &
                        mae_miss$parameter == param &
                        mae_miss$method == m]
    cat(sprintf("%s / %s: MAE 0% -> 50% : %.4f -> %.4f (%.2fx)\n",
                param, m, base, high, high / base))
  }
}

```

```

## Intercept / GHK: MAE 0% -> 50% : 0.1564 -> 0.1709 (1.09x)
## Intercept / GQT: MAE 0% -> 50% : 0.1585 -> 0.1689 (1.07x)
## range / GHK: MAE 0% -> 50% : 0.0678 -> 0.1083 (1.60x)
## range / GQT: MAE 0% -> 50% : 0.0660 -> 0.1066 (1.62x)

```

5.3 Runtime by missingness rate

Smaller effective sample sizes mean fewer likelihood evaluations, so runtime should decrease with the missingness rate. The report cites the GHK 0% vs 50% comparison directly.

```

rt_miss <- unique(long_miss[long_miss$converged & long_miss$parameter == "Intercept",
                           c("scenario_id", "replicate", "method",
                             "missing_rate", "runtime_sec")])
runtime_miss <- aggregate(runtime_sec ~ missing_rate + method, data = rt_miss, FUN = mean)
runtime_miss_wide <- reshape(runtime_miss, idvar = "missing_rate",
                             timevar = "method", direction = "wide")
print(runtime_miss_wide, row.names = FALSE, digits = 3)

## missing_rate runtime_sec.GHK runtime_sec.GQT
##          0.0          0.742          0.0932
##          0.1          0.642          0.0813
##          0.2          0.570          0.0666
##          0.3          0.447          0.0564
##          0.5          0.327          0.0358

save_table(runtime_miss_wide, "missing_runtime")

```

5.4 Figure: MAE vs missingness rate

```

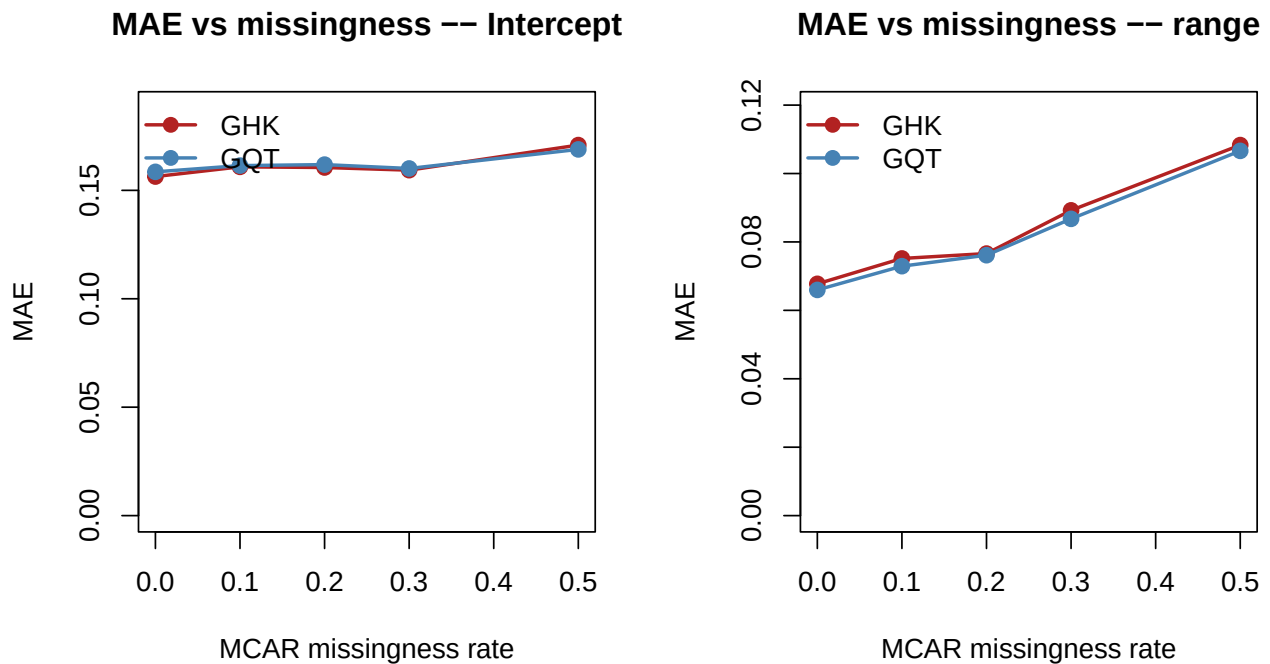
plot_missing <- function() {
  par(mfrow = c(1, 2))
  for (param in c("Intercept", "range")) {
    d <- mae_miss[mae_miss$parameter == param, ]
    y_max <- max(d$MAE) * 1.1
    plot(NA, xlim = c(0, 0.5), ylim = c(0, y_max),
         xlab = "MCAR missingness rate", ylab = "MAE",
         main = sprintf("MAE vs missingness -- %s", param))
    for (m in c("GHK", "GQT")) {

```

```

sub <- d[d$method == m, ]
sub <- sub[order(sub$missing_rate), ]
col <- ifelse(m == "GHK", "firebrick", "steelblue")
lines(sub$missing_rate, sub$MAE, col = col, lwd = 2)
points(sub$missing_rate, sub$MAE, col = col, pch = 19, cex = 1.2)
}
legend("topleft", legend = c("GHK", "GQT"),
      col = c("firebrick", "steelblue"), lwd = 2, pch = 19, bty = "n")
}
par(mfrow = c(1, 1))
}
plot_missing()

```



```

pdf("visualization/report/missing_mae_vs_rate.pdf", width = 10, height = 4.5)
plot_missing()
dev.off()

```

```

## pdf
## 2

```

6 Study 4: IRREGULAR (Spatial Layout)

Three layouts at $n = 49$: regular 7×7 grid, uniform random, and four Gaussian clusters. Same generating parameters as MISSING: Poisson(5), $r = 0.3$, 200 replications per layout.

6.1 Load and tidy

```

DATASET_IRR <- "IRREGULAR"
ids_irr <- list_scenarios(DATASET_IRR, dir = "data/generated")
fits_irr <- lapply(ids_irr, function(id) load_fit_result(DATASET_IRR, id))
names(fits_irr) <- ids_irr

```

```

extra_irr <- function(cfg) data.frame(
  layout_type = cfg$layout$layout_type,
  stringsAsFactors = FALSE
)

long_irr <- do.call(rbind, lapply(fits_irr, tidy_fit, extra_cols = extra_irr))
long_irr$truth <- NA_real_
long_irr$truth[long_irr$parameter %in% "Intercept"] <- log(5)
long_irr$truth[long_irr$parameter %in% "range"] <- 0.3
long_irr$abs_err <- abs(long_irr$estimate - long_irr$truth)
long_irr$err <- long_irr$estimate - long_irr$truth

```

6.2 MAE and bias by layout

```

mae_irr <- aggregate(
  abs_err ~ scenario_id + method + parameter,
  data = long_irr[long_irr$converged, ],
  FUN = mean
)
names(mae_irr)[4] <- "MAE"
mae_irr <- mae_irr[order(mae_irr$parameter, mae_irr$scenario_id,
  mae_irr$method), ]
print(mae_irr, row.names = FALSE, digits = 3)

```

```

## scenario_id method parameter    MAE
## clustered   GHK Intercept 0.2049
## clustered   GQT Intercept 0.1998
## grid        GHK Intercept 0.1422
## grid        GQT Intercept 0.1399
## random      GHK Intercept 0.1510
## random      GQT Intercept 0.1503
## clustered   GHK range 0.0605
## clustered   GQT range 0.0581
## grid        GHK range 0.0639
## grid        GQT range 0.0627
## random      GHK range 0.0638
## random      GQT range 0.0610

```

```

save_table(mae_irr, "irregular_mae")

```

```

bias_irr <- aggregate(
  err ~ scenario_id + method + parameter,
  data = long_irr[long_irr$converged, ],
  FUN = mean
)
names(bias_irr)[4] <- "bias"
print(bias_irr, row.names = FALSE, digits = 3)

```

```

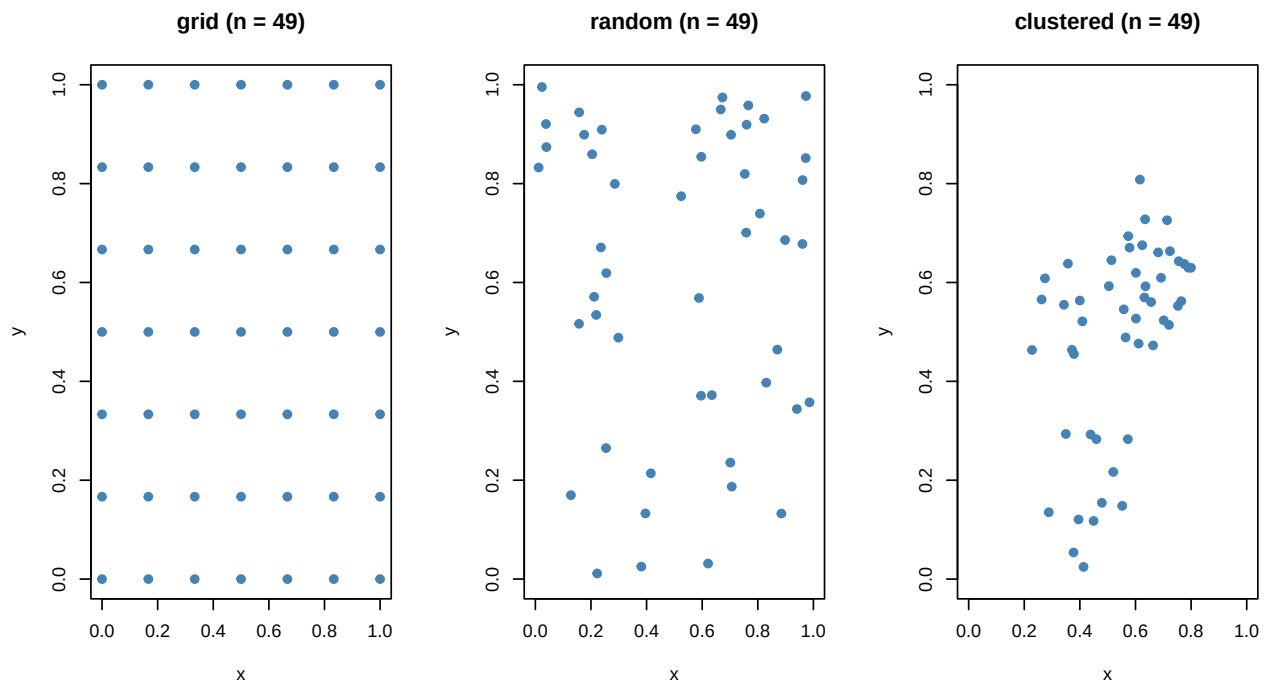
## scenario_id method parameter    bias
## clustered   GHK Intercept -0.021865
## grid        GHK Intercept -0.000108
## random      GHK Intercept  0.001421
## clustered   GQT Intercept -0.008770
## grid        GQT Intercept  0.001121

```

```
##      random    GQT Intercept 0.003071
##      clustered GHK      range -0.007117
##      grid      GHK      range 0.005200
##      random    GHK      range 0.005328
##      clustered GQT      range -0.021346
##      grid      GQT      range 0.009486
##      random    GQT      range 0.003170
```

6.3 Figure: layouts side-by-side, then MAE bars

```
plot_layouts <- function() {
  par(mfrow = c(1, 3))
  for (id in c("grid", "random", "clustered")) {
    scen <- load_scenario(DATASET_IRR, id, dir = "data/generated")
    plot(scen$locations[, "x"], scen$locations[, "y"],
         pch = 19, cex = 1.0, col = "steelblue",
         xlab = "x", ylab = "y", xlim = c(0, 1), ylim = c(0, 1),
         main = sprintf("%s (n = %d)", id, nrow(scen$locations)))
  }
  par(mfrow = c(1, 1))
}
plot_layouts()
```



```
pdf("visualization/report/irregular_layouts.pdf", width = 10, height = 3.5)
plot_layouts()
dev.off()
```

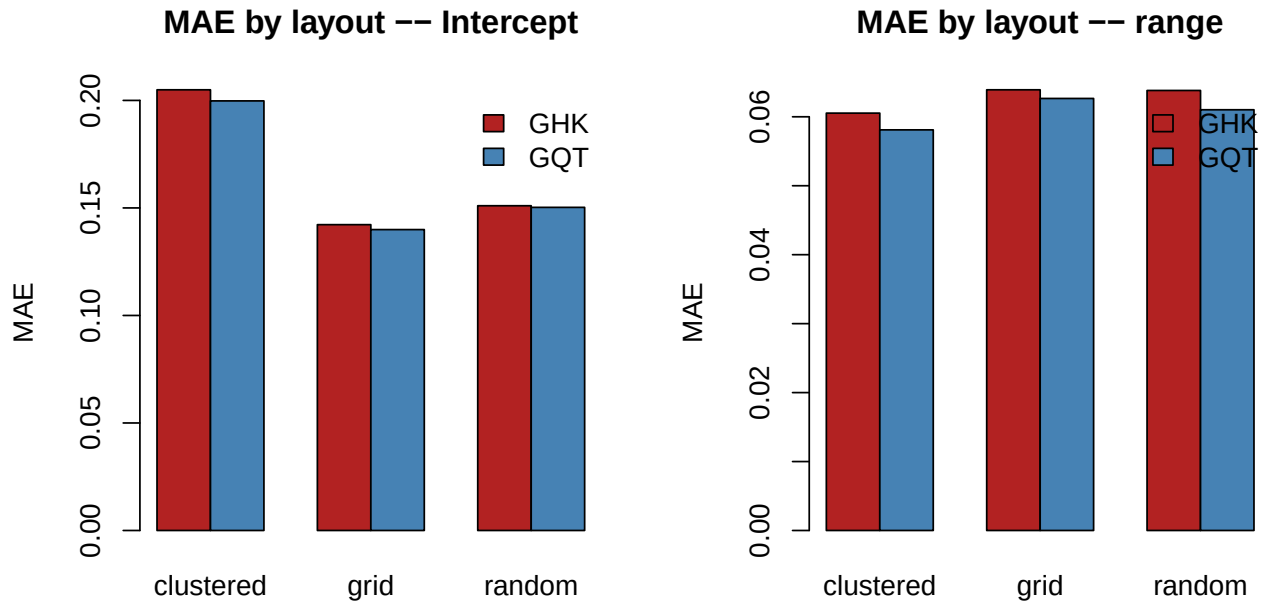
```
## pdf
## 2
```

```
plot_irr_mae <- function() {
  par(mfrow = c(1, 2))
  for (param in c("Intercept", "range")) {
```

```

d <- mae_irr[mae_irr$parameter == param, ]
layouts_order <- c("grid", "random", "clustered")
d$scenario_id <- factor(d$scenario_id, levels = layouts_order)
wide <- reshape(d[, c("scenario_id", "method", "MAE")],
               idvar = "scenario_id", timevar = "method",
               direction = "wide")
mat <- rbind(wide$MAE.GHK, wide$MAE.GQT)
colnames(mat) <- wide$scenario_id
rownames(mat) <- c("GHK", "GQT")
barplot(mat, beside = TRUE,
        col = c("firebrick", "steelblue"),
        ylab = "MAE",
        main = sprintf("MAE by layout -- %s", param),
        legend.text = rownames(mat),
        args.legend = list(x = "topright", bty = "n"))
}
par(mfrow = c(1, 1))
}
plot_irr_mae()

```



```

pdf("visualization/report/irregular_mae.pdf", width = 10, height = 4.5)
plot_irr_mae()
dev.off()

```

```

## pdf
## 2

```

7 Study 5: ZIP (Zero-Inflation)

Three zero-inflation levels $p_0 \in \{0, 0.2, 0.4\}$, each fitted both correctly (ZIP) and incorrectly (Poisson). $n = 49$, $r = 0.3$.

7.1 Load both well-specified and misspecified fits

```
DATASET_ZIP <- "ZIP"
inv_zip <- list_fit_results(DATASET_ZIP, dir = "data/fits")

fits_zip <- lapply(seq_len(nrow(inv_zip)), function(i) {
  load_fit_result(
    DATASET_ZIP,
    inv_zip$scenario_id[i],
    misspec_family = if (is.na(inv_zip$misspec_family[i])) NULL
                      else inv_zip$misspec_family[i]
  )
})
names(fits_zip) <- paste(
  inv_zip$scenario_id,
  ifelse(is.na(inv_zip$misspec_family), "wellspec", inv_zip$misspec_family),
  sep = "__"
)

# Custom tidier because ZIP needs both p0 and the actual fit family.
tidy_zip <- function(f) {
  r <- f$results
  cfg <- f$scenario_config
  rows <- list()
  for (i in seq_len(nrow(r))) {
    if (!isTRUE(r$converged[i])) next
    est <- r$estimates[[i]]
    if (is.null(est) || length(est) == 0) next
    for (pname in names(est)) {
      rows[[length(rows) + 1]] <- data.frame(
        scenario_id = f$scenario_id,
        p0          = cfg$marginal$od, # ZIP probability stored under `od`
        fit_family   = f$fit_marginal_spec$family,
        replicate    = r$replicate[i],
        method       = r$method[i],
        parameter     = pname,
        estimate      = as.numeric(est[[pname]]),
        stringsAsFactors = FALSE
      )
    }
  }
  do.call(rbind, rows)
}

long_zip <- do.call(rbind, lapply(fits_zip, tidy_zip))
long_zip$truth <- NA_real_
long_zip$truth[long_zip$parameter %in% "Intercept"] <- log(5)
long_zip$truth[long_zip$parameter %in% "range"] <- 0.3
long_zip$abs_err <- abs(long_zip$estimate - long_zip$truth)
```

7.2 Convergence

GQT-ZIP shows mild convergence drops at $p_0 \in \{0.2, 0.4\}$. This is the first sign of the GQT instability under strong zero-inflation, which the MAE comparison and well-specified-GQT range error make explicit.

```

# Reconstruct one-row-per-fit for convergence purposes (the long
# table dropped failed rows during tidying).
conv_zip_rows <- do.call(rbind, lapply(fits_zip, function(f) {
  cfg <- f$scenario_config
  data.frame(
    p0 = cfg$marginal$od,
    fit_family = f$fit_marginal_spec$family,
    method = f$results$method,
    converged = f$results$converged,
    stringsAsFactors = FALSE
  )
}))
conv_zip <- aggregate(converged ~ p0 + fit_family + method,
  data = conv_zip_rows, FUN = mean)
conv_zip <- conv_zip[order(conv_zip$fit_family, conv_zip$p0, conv_zip$method), ]
print(conv_zip, row.names = FALSE, digits = 4)

##   p0 fit_family method converged
## 0.0 poisson GHK 1.000
## 0.0 poisson GQT 1.000
## 0.2 poisson GHK 1.000
## 0.2 poisson GQT 1.000
## 0.4 poisson GHK 1.000
## 0.4 poisson GQT 1.000
## 0.0 zip GHK 1.000
## 0.0 zip GQT 1.000
## 0.2 zip GHK 1.000
## 0.2 zip GQT 0.995
## 0.4 zip GHK 1.000
## 0.4 zip GQT 0.995

save_table(conv_zip, "zip_convergence")

```

7.3 MAE: well-specified vs misspecified, side by side

```

mae_zip <- aggregate(
  abs_err ~ p0 + fit_family + method + parameter,
  data = long_zip[!is.na(long_zip$abs_err), ],
  FUN = mean
)
names(mae_zip)[5] <- "MAE"

mae_zip_wide <- reshape(
  mae_zip, idvar = c("p0", "method", "parameter"),
  timevar = "fit_family", direction = "wide"
)
mae_zip_wide$pct <- 100 * (mae_zip_wide$MAE.poisson - mae_zip_wide$MAE.zip) /
  mae_zip_wide$MAE.zip
mae_zip_wide <- mae_zip_wide[order(mae_zip_wide$parameter,
  mae_zip_wide$p0,
  mae_zip_wide$method), ]
print(mae_zip_wide, row.names = FALSE, digits = 3)

##   p0 method parameter MAE.poisson MAE.zip   pct

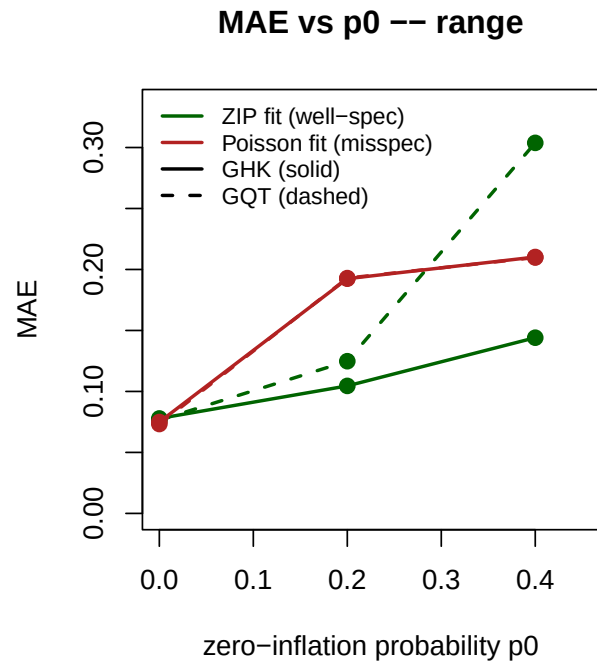
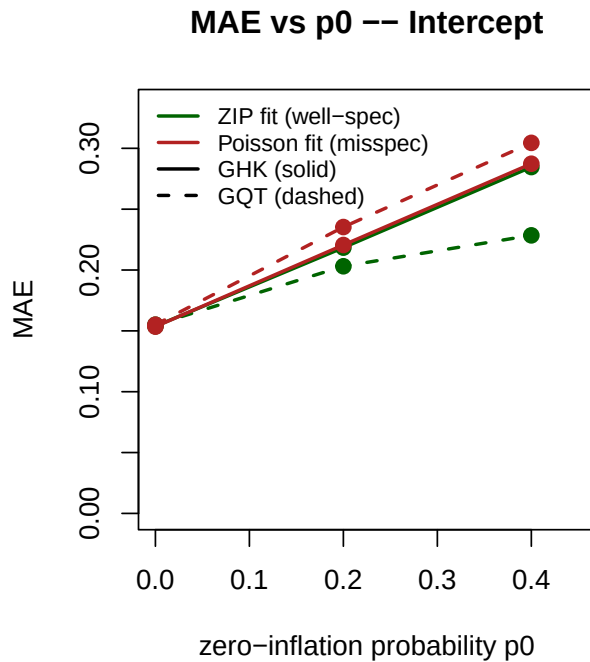
```

```
## 0.0 GHK Intercept 0.1535 0.1538 -0.2249
## 0.0 GQT Intercept 0.1551 0.1550 0.0146
## 0.2 GHK Intercept 0.2206 0.2184 0.9941
## 0.2 GQT Intercept 0.2353 0.2031 15.8821
## 0.4 GHK Intercept 0.2873 0.2846 0.9574
## 0.4 GQT Intercept 0.3045 0.2285 33.2542
## 0.0 GHK range 0.0749 0.0779 -3.7517
## 0.0 GQT range 0.0733 0.0764 -4.0692
## 0.2 GHK range 0.1924 0.1045 84.0633
## 0.2 GQT range 0.1929 0.1248 54.5949
## 0.4 GHK range 0.2102 0.1441 45.9307
## 0.4 GQT range 0.2098 0.3038 -30.9432
```

```
save_table(mae_zip_wide, "zip_mae_wide")
```

7.4 Figure: MAE vs zero-inflation level

```
plot_zip <- function() {
  par(mfrow = c(1, 2))
  for (param in c("Intercept", "range")) {
    d <- mae_zip[mae_zip$parameter == param, ]
    y_max <- max(d$MAE) * 1.1
    plot(NA, xlim = c(0, 0.45), ylim = c(0, y_max),
         xlab = "zero-inflation probability p0",
         ylab = "MAE",
         main = sprintf("MAE vs p0 -- %s", param))
    for (ff in c("zip", "poisson")) {
      for (m in c("GHK", "GQT")) {
        sub <- d[d$fit_family == ff & d$method == m, ]
        sub <- sub[order(sub$p0), ]
        col <- if (ff == "zip") "darkgreen" else "firebrick"
        lty <- if (m == "GHK") 1 else 2
        lines(sub$p0, sub$MAE, col = col, lty = lty, lwd = 2)
        points(sub$p0, sub$MAE, col = col, pch = 19, cex = 1.2)
      }
    }
    legend("topleft",
          legend = c("ZIP fit (well-spec)", "Poisson fit (misspec)",
                    "GHK (solid)", "GQT (dashed)"),
          col = c("darkgreen", "firebrick", "black", "black"),
          lty = c(1, 1, 1, 2), lwd = 2, bty = "n", cex = 0.8)
  }
  par(mfrow = c(1, 1))
}
plot_zip()
```



```
pdf("visualization/report/zip_mae_vs_p0.pdf", width = 10, height = 4.5)
plot_zip()
dev.off()
```

```
## pdf
## 2
```

7.5 GQT instability at $p_0 = 0.4$

The most striking single finding: even when ZIP is well-specified, GQT inflates range MAE at $p_0 = 0.4$ to roughly twice GHK's value.

```
zip_well_p04 <- mae_zip[mae_zip$fit_family == "zip" &
  mae_zip$p0 == 0.4 &
  mae_zip$parameter == "range", ]
print(zip_well_p04, row.names = FALSE, digits = 3)
```

```
##   p0 fit_family method parameter    MAE
## 0.4      zip      GHK      range 0.144
## 0.4      zip      GQT      range 0.304
```

8 Headline Numbers for the Report

This block consolidates every numerical claim cited in the report's written sections. Each line corresponds to a specific sentence in the report. If the underlying data changes, this block reflects it immediately.

```
cat("==== Headline numbers =====\n\n")
```

```
## ===== Headline numbers =====
```

```
# --- K13A-GRID ---
```

```
cat("--- K13A-GRID ---\n")
```

```
## --- K13A-GRID ---
```

```

cat(sprintf("Convergence: GHK %.4f, GQT %.4f\n",
            conv_k13a$converged[conv_k13a$method == "GHK"],
            conv_k13a$converged[conv_k13a$method == "GQT"]))

## Convergence: GHK 1.0000, GQT 0.9994

cat(sprintf("GQT speedup at n=25: %.2fx, n=49: %.2fx\n",
            runtime_wide$ratio_GHK_to_GQT[runtime_wide$n_total == 25],
            runtime_wide$ratio_GHK_to_GQT[runtime_wide$n_total == 49]))

## GQT speedup at n=25: 8.56x, n=49: 8.25x

cat(sprintf("Max |MAE %% diff| GHK vs GQT across cells: %.1f%%\n\n",
            max(abs(mae_k13a_wide$pct_diff), na.rm = TRUE)))

## Max |MAE % diff| GHK vs GQT across cells: 33.4%

# --- MISSPEC ---
cat("--- MISSPEC ---\n")

## --- MISSPEC ---

get_mae <- function(cell, m, p) {
  v <- mae_misspec$MAE[mae_misspec$cell_label == cell &
                      mae_misspec$method == m &
                      mae_misspec$parameter == p]
  if (length(v) == 0) NA else v
}

cat(sprintf("Range MAE NB->NB GHK: %.3f; NB->Poisson GHK: %.3f (%.2fx)\n",
            get_mae("negbin -> negbin", "GHK", "range"),
            get_mae("negbin -> poisson", "GHK", "range"),
            get_mae("negbin -> poisson", "GHK", "range") /
            get_mae("negbin -> negbin", "GHK", "range"))))

## Range MAE NB->NB GHK: 0.105; NB->Poisson GHK: 0.210 (2.00x)

cat(sprintf("Mean estimated range under NB->Poisson (GHK): %.3f (truth 0.3)\n",
            mean(long_misspec$estimate[long_misspec$cell_label == "negbin -> poisson" &
                                       long_misspec$method == "GHK" &
                                       long_misspec$parameter == "range"])))

## Mean estimated range under NB->Poisson (GHK): 0.090 (truth 0.3)

cat(sprintf("Mean GQT log-likelihood under NB->Poisson: %.2e\n\n",
            mean(long_misspec$loglik[long_misspec$cell_label == "negbin -> poisson" &
                                       long_misspec$method == "GQT" &
                                       long_misspec$parameter == "Intercept"])))

## Mean GQT log-likelihood under NB->Poisson: -3.18e+13

# --- MISSING ---
cat("--- MISSING ---\n")

## --- MISSING ---

for (param in c("Intercept", "range")) {
  for (m in c("GHK", "GQT")) {
    base <- mae_miss$MAE[mae_miss$missing_rate == 0 &
                        mae_miss$parameter == param &
                        mae_miss$method == m]
  }
}

```

```

    high <- mae_miss$MAE[mae_miss$missing_rate == 0.5 &
                        mae_miss$parameter == param &
                        mae_miss$method == m]
    cat(sprintf(" %s/%s : 0%% -> 50%% %.4f -> %.4f (%.2fx)\n",
                param, m, base, high, high / base))
  }
}

## Intercept/GHK : 0% -> 50% 0.1564 -> 0.1709 (1.09x)
## Intercept/GQT : 0% -> 50% 0.1585 -> 0.1689 (1.07x)
## range/GHK : 0% -> 50% 0.0678 -> 0.1083 (1.60x)
## range/GQT : 0% -> 50% 0.0660 -> 0.1066 (1.62x)

# Bias support: largest |bias| across all rates
cat(sprintf(" Max |bias|: Intercept %.4f; range %.4f (across all rates, both methods)\n",
            max(abs(bias_miss$bias[bias_miss$parameter == "Intercept"])),
            max(abs(bias_miss$bias[bias_miss$parameter == "range"]))))

## Max |bias|: Intercept 0.0235; range 0.0057 (across all rates, both methods)

# Runtime support
cat(sprintf(" Runtime GHK 0% -> 50%: %.3f s -> %.3f s\n",
            runtime_miss_wide$runtime_sec.GHK[runtime_miss_wide$missing_rate == 0],
            runtime_miss_wide$runtime_sec.GHK[runtime_miss_wide$missing_rate == 0.5]))

## Runtime GHK 0% -> 50%: 0.742 s -> 0.327 s

cat("\n")

# --- IRREGULAR ---
cat("--- IRREGULAR ---\n")

## --- IRREGULAR ---

get_irr <- function(layout, m, p) {
  v <- mae_irr$MAE[mae_irr$scenario_id == layout &
                  mae_irr$method == m &
                  mae_irr$parameter == p]
  if (length(v) == 0) NA else v
}

cat(sprintf("Intercept MAE grid: %.3f; clustered: %.3f (+%.0f%%)\n",
            get_irr("grid", "GHK", "Intercept"),
            get_irr("clustered", "GHK", "Intercept"),
            100 * (get_irr("clustered", "GHK", "Intercept") /
                  get_irr("grid", "GHK", "Intercept") - 1)))

## Intercept MAE grid: 0.142; clustered: 0.205 (+44%)

cat(sprintf("Range MAE grid: %.3f; clustered: %.3f\n",
            get_irr("grid", "GHK", "range"),
            get_irr("clustered", "GHK", "range")))

## Range MAE grid: 0.064; clustered: 0.061

# --- ZIP ---
cat("--- ZIP ---\n")

## --- ZIP ---

```

```

get_zip <- function(p0v, ff, m, p) {
  v <- mae_zip$MAE[mae_zip$p0 == p0v &
    mae_zip$fit_family == ff &
    mae_zip$method == m &
    mae_zip$parameter == p]
  if (length(v) == 0) NA else v
}
cat(sprintf("Range MAE p0=0.2 ZIP fit GHK: %.3f; Poisson fit GHK: %.3f (+%.0f%%)\n",
  get_zip(0.2, "zip", "GHK", "range"),
  get_zip(0.2, "poisson", "GHK", "range"),
  100 * (get_zip(0.2, "poisson", "GHK", "range") /
    get_zip(0.2, "zip", "GHK", "range") - 1)))

## Range MAE p0=0.2 ZIP fit GHK: 0.105; Poisson fit GHK: 0.192 (+84%)

cat(sprintf("Range MAE p0=0.4 ZIP-GHK: %.3f; ZIP-GQT: %.3f (GQT instability)\n",
  get_zip(0.4, "zip", "GHK", "range"),
  get_zip(0.4, "zip", "GQT", "range")))

## Range MAE p0=0.4 ZIP-GHK: 0.144; ZIP-GQT: 0.304 (GQT instability)

# Convergence support for the GQT instability story
cat(sprintf("Min ZIP-GQT convergence: %.4f at p0=%.1f\n",
  min(conv_zip$converged[conv_zip$fit_family == "zip" &
    conv_zip$method == "GQT"]),
  conv_zip$p0[conv_zip$fit_family == "zip" &
    conv_zip$method == "GQT" &
    conv_zip$converged ==
    min(conv_zip$converged[conv_zip$fit_family == "zip" &
      conv_zip$method == "GQT"])[1]))

## Min ZIP-GQT convergence: 0.9950 at p0=0.2

```

9 Files Produced

All tables saved as CSV under `tables/report/` and figures saved as PDF under `visualization/report/`. The naming convention matches the references in the LaTeX report:

- `tables/report/k13a_runtime.csv`, `tables/report/k13a_mae_wide.csv`, `tables/report/k13a_bias.csv`
- `tables/report/misspec_mae.csv`
- `tables/report/missing_mae.csv`, `tables/report/missing_bias.csv`, `tables/report/missing_runtime.csv`
- `tables/report/irregular_mae.csv`
- `tables/report/zip_mae_wide.csv`, `tables/report/zip_convergence.csv`
- `visualization/report/k13a_mae_intercept.pdf`, `k13a_mae_range.pdf`
- `visualization/report/misspec_estimates.pdf`
- `visualization/report/missing_mae_vs_rate.pdf`
- `visualization/report/irregular_layouts.pdf`, `irregular_mae.pdf`
- `visualization/report/zip_mae_vs_p0.pdf`

10 Session Info

```

sessionInfo()

## R version 4.1.1 (2021-08-10)

```

```

## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 20.04.3 LTS
##
## Matrix products: default
## BLAS/LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.8.so
##
## locale:
## [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
## [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
## [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=C
## [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
## [9] LC_ADDRESS=C             LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## attached base packages:
## [1] stats      graphics  grDevices datasets  utils      methods    base
##
## other attached packages:
## [1] ggplot2_3.3.5 gcKrig_1.1.8
##
## loaded via a namespace (and not attached):
## [1] Rcpp_1.1.1      pillar_1.6.4    compiler_4.1.1  tools_4.1.1
## [5] digest_0.6.28  evaluate_1.0.5  lifecycle_1.0.1 tibble_3.1.5
## [9] gtable_0.3.0   pkgconfig_2.0.3 rlang_1.2.0     cli_3.1.0
## [13] yaml_2.2.1     xfun_0.57       fastmap_1.2.0   withr_2.4.2
## [17] dplyr_1.0.7    knitr_1.51      generics_0.1.1  vctrs_0.3.8
## [21] grid_4.1.1     tidyselect_1.1.1 glue_1.4.2      R6_2.5.1
## [25] fansi_0.5.0    rmarkdown_2.31  farver_2.1.0    purrr_0.3.4
## [29] magrittr_2.0.1 scales_1.1.1    ellipsis_0.3.2  htmltools_0.5.9
## [33] colorspace_2.0-2 renv_1.2.2      labeling_0.4.2  tinytex_0.59
## [37] utf8_1.2.2     munsell_0.5.0   crayon_1.4.1

```